# Granular Instrumental Variables

**Gabaix and Koijen (2022)**

Roy Kisluk

May 19, 2026

# Aggregate shocks confound firm effects

**The Goal:** Estimate causal elasticity of wages to labor demand

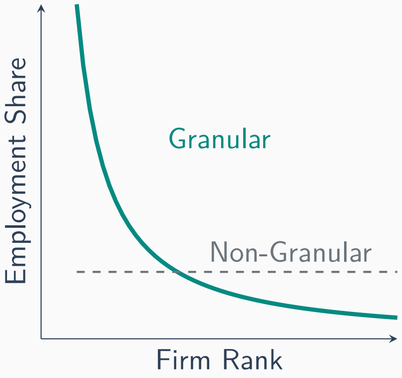
The Identification Challenge: **Simultaneity**

Quantities and prices are jointly determined in equilibrium.

# LLN fails in granular economies

1. In a standard economy, micro-shocks cancel out at a rate of $1/\sqrt{N}$ ▸ Proof

2. In a **granular** economy, size distribution of firms follows a power law: $size = \frac{1}{rank^{\frac{1}{\alpha}}}$
   - **Zipf's Law**: $\alpha = 1 \implies \text{size} \propto \frac{1}{\text{rank}}$. First firm is x2 the second, x3 the third, and so on.
   - **Israel**: $\alpha = 1.2$. First firm is x1.8 the second, and x2.6 the third.
   - Large firms' idiosyncratic shocks drive aggregate outcomes



# A Simple Model

# Simultaneity prevents OLS estimation

Economy with $N$ firms. System is given by:

$$w_t = \psi L_{St} + \varepsilon_t \qquad \text{(Aggregate Supply)}$$

$$L_{it} = \phi^d w_t + \eta_t + u_{it} \qquad \text{(Firm-specific Demand)}$$

Where:

- $w_t$ - wage (observed)
- $L_{it}$ - number of workers at firm $i$ (observed) Aggregation
- $\eta_t$ and $\varepsilon_t$ - aggregate shocks (unobserved)
- $u_{it}$ - idiosyncratic shock to firm $i$ (unobserved)
- Main assumption: $corr(u_{it}, \eta_t) = 0,\ corr(u_{it}, \varepsilon_t) = 0$ Assumptions

We cannot estimate $\psi$ (labor supply elasticity) and $\phi^d$ (labor demand elasticity) by OLS, due to simultaneity.

# GIV differences size and equal weights

GIV exploits difference between size-weighted and equal-weighted averages.

- Define equal-weighted average:

$$L_{Et} = \frac{1}{N} \sum_{i=1}^{N} L_{it}$$

- Define size-weighted average:

$$L_{St} = \sum_{i=1}^{N} S_i L_{it}, \quad \sum_{i=1}^{N} S_i = 1$$

GIV is the difference between them:

$$z_t := L_{\Gamma t} = L_{St} - L_{Et} \tag{GIV}$$

# GIV isolates idiosyncratic shocks

Substitute demand equation into averages: ▸ Proof

$$L_{St} = \phi^d w_t + \eta_t + u_{St}$$
$$L_{Et} = \phi^d w_t + \eta_t + u_{Et}$$

Common components ($\phi^d w_t + \eta_t$) difference out:

$$z_t = u_{St} - u_{Et} = \sum_{i=1}^{N} S_i u_{it} - \frac{1}{N}\sum_{i=1}^{N} u_{it} = \sum_{i=1}^{N}(S_i - \frac{1}{N})u_{it} \qquad \text{(GIV)}$$

$z_t$ is made *only* of idiosyncratic shocks.

# GIV enables 2SLS estimation

**IV Estimation:**

- **First Stage:** Regress $L_{St}$ on $z_t$. Extracts $\widehat{L}_{St}$, the labor demand variation driven *only* by idiosyncratic shocks.
- **Second Stage:** Regress $w_t$ on $\widehat{L}_{St}$ to obtain unbiased $\psi$.

**Fundamental IV assumptions:**

- **Relevance** (Cov$(L_{St}, z_t) \neq 0$):
  Idiosyncratic granular shocks must move aggregate market. This requires a concentrated market (e.g., high Herfindahl index).
- **Exclusion Restriction** (Cov$(z_t, \varepsilon_t) = 0$, Cov$(z_t, \eta_t) = 0$):
  Idiosyncratic granular shocks must not directly affect aggregate wages through any channel, i.e. they are orthogonal to macro shocks ($z_t \perp \eta_t, \varepsilon_t$).

# GIV yields both elasticities

Using supply equation, this is what we get from the second stage:

$$\mathbb{E}[z_t \varepsilon_t] = 0, \quad w_t = \psi L_{St} + \varepsilon_t \implies \mathbb{E}[(w_t - \psi L_{St}) z_t] = 0 \implies \psi = \frac{\mathbb{E}[w_t z_t]}{\mathbb{E}[L_{St} z_t]}$$

Using equal-weighted demand equation: ▸ Proof

$$\mathbb{E}[(L_{Et} - \phi^d w_t) z_t] = 0 \implies \phi^d = \frac{\mathbb{E}[L_{Et} z_t]}{\mathbb{E}[w_t z_t]}$$

# The Full Model

Handling heterogeneous factor loadings

# Heterogeneous loadings bias simple GIV

We now allow for heterogeneous loadings $\lambda_i$:

$$L_{it} = \phi w_t + \lambda_i \eta_t + u_{it}$$

The simple GIV assumes every firm has the same loading $\lambda_i = 1$.

If size is correlated with the macro shock, the simple GIV becomes: ▸ Proof

$$z_t = (\lambda_S - \lambda_E)\eta_t + (u_{St} - u_{Et})$$

If $\lambda_S \neq \lambda_E$, the instrument is **invalid** ($\mathbb{E}[z_t \eta_t] \neq 0$, i.e., the exclusion restriction is violated).

# Orthogonal weights restore exclusion

Substitute demand equation into definition of weighted GIV:

$$
\begin{aligned}
z_t &= \sum \Gamma_i L_{it} = \sum \Gamma_i (\phi w_t + \lambda_i \eta_t + u_{it}) \\
&= \phi w_t \left(\sum \Gamma_i\right) + \eta_t \left(\sum \Gamma_i \lambda_i\right) + \sum \Gamma_i u_{it}
\end{aligned}
$$

Where $\Gamma$ is the vector of weights that we want to find.

To ensure $z_t$ contains *only* idiosyncratic shocks ($u_{it}$), we must satisfy two conditions:

1. $\sum \Gamma_i = 0$
2. $\sum \Gamma_i \lambda_i = 0$

# Projection finds optimal weights

For this, we use $N \times N$ projection matrix $Q$, orthogonal to factor loadings ($Q\lambda = 0$):

▸ Refresher

$$Q = I - \lambda(\lambda'\lambda)^{-1}\lambda'$$

Optimal weights are then given by $\Gamma' = S'Q$.

- Intuitively, we want $\Gamma$ to be as close as possible to $S$ but satisfy $\Gamma \perp \lambda$
- This is a standard projection problem: $\Gamma = S - \lambda(\lambda'\lambda)^{-1}\lambda' S$

▸ Propositions

# Application

GIV in the Israeli Labor Market

# Israeli tech is highly granular

Israeli high-tech industry is dominated by a few multinational R&D centers.

- Largest firms account for disproportionate share of employment: top $10 = 15\%$, top $50 = 30\%$
- Size distribution of tech in Israel exhibits strong power law:

$$\frac{Q(\bar{q_i})}{Q(\bar{q_1})} = \frac{1}{i^{\frac{1}{\zeta}}}, \quad \zeta \approx 1.2$$

- Difficult to separate Israeli macro shocks (wars, interest rates) from hiring dynamics

# GIV unconfounds tech spillovers

# Appendix A

Derivations

# Aggregate Demand

Let $\bar{E}$ be the steady-state aggregate employment. Firm $i$'s labor demand in levels is given by:

$$E_{it} = \bar{E} S_i (1 + L_{it}), \quad \text{with} \quad L_{it} = \phi^d w_t + \eta_t + u_{it}$$

where $S_i$ is the firm's steady-state employment share ($\sum_{i=1}^N S_i = 1$), and $L_{it}$ is the fractional change in employment.
Aggregate labor demand in levels $(E_t)$ is the sum of firm-level demands:

$$\begin{aligned} E_t &= \sum_{i=1}^N E_{it} = \sum_{i=1}^N \bar{E} S_i (1 + L_{it}) \\ &= \bar{E} \sum_{i=1}^N S_i + \bar{E} \sum_{i=1}^N S_i L_{it} \end{aligned}$$

Defining the size-weighted average $L_{St} := \sum_{i=1}^N S_i L_{it}$, and since $\sum_{i=1}^N S_i = 1$, we get:

$$E_t = \bar{E}(1 + L_{St})$$

# Assumptions

## The Central Identification Assumption

Individual shocks $u_{it}$ are truly idiosyncratic—uncorrelated with aggregate shocks $(\eta_t, \epsilon_t)$ and controls $(C_t^y, C_t^p)$:

$$u_t \perp \eta_t, \epsilon_t, C_t^y, C_t^p \tag{1}$$

### Baseline Simplifications

- A1 Factor loadings $\lambda_i$ are known (relaxed in §IV.B, estimated using GMM / PCA).
- A2 Controls have factor structure: $C_{it}^y = \lambda_i c_t^y$.

### Error Structure

- A3 Homoskedastic and uncorrelated across $i$: $\mathbb{E}[u_t u_t'] = \sigma_u^2 I$.
- A4 General heteroskedasticity $V^u$ (relaxed for optimality).

*Note: All random variables are assumed to have finite fourth moments and are i.i.d. across periods.*

Back

# Equal weights decay at $1/\sqrt{N}$

Suppose there are $N$ firms, each receiving an independent idiosyncratic shock $u_{it}$. Assume these shocks have mean 0 and a common variance $\sigma^2$:

$$\mathbb{E}[u_{it}] = 0, \quad \text{Var}(u_{it}) = \sigma^2, \quad \text{Cov}(u_{it}, u_{jt}) = 0 \text{ for } i \neq j$$

In a non-granular economy, the aggregate shock is an **equal-weighted** average:

$$u_{Et} = \frac{1}{N} \sum_{i=1}^{N} u_{it}$$

The variance of the aggregate shock is:

$$\text{Var}(u_{Et}) = \text{Var}\left(\frac{1}{N} \sum u_{it}\right) = \frac{1}{N^2} \sum \text{Var}(u_{it}) = \frac{N\sigma^2}{N^2} = \frac{\sigma^2}{N}$$

The aggregate volatility is therefore: $\sigma_{agg} = \frac{\sigma}{\sqrt{N}}$

Back

# Size-weighting aggregates shocks

The size-weighted outcome is defined as $L_{St} = \sum_{i=1}^{N} S_i L_{it}$, where $S_i$ are weights such that $\sum S_i = 1$.

$$L_{St} = \sum_{i=1}^{N} S_i(\phi^d w_t + \eta_t + u_{it})$$

$$L_{St} = \sum_{i=1}^{N} S_i \phi^d w_t + \sum_{i=1}^{N} S_i \eta_t + \sum_{i=1}^{N} S_i u_{it}$$

Since $\phi^d$, $w_t$, and $\eta_t$ do not depend on the index $i$, we can pull them out:

$$L_{St} = \phi^d w_t \left( \sum_{i=1}^{N} S_i \right) + \eta_t \left( \sum_{i=1}^{N} S_i \right) + \sum_{i=1}^{N} S_i u_{it}$$

Using the property $\sum S_i = 1$ and defining the size-weighted idiosyncratic shock as $u_{St} = \sum S_i u_{it}$, we get:

$$L_{St} = \phi^d w_t + \eta_t + u_{St}$$



◂ Back

# Equal-weighting averages shocks

The equal-weighted outcome is defined as $L_{Et} = \frac{1}{N}\sum_{i=1}^{N} L_{it}$.

$$L_{Et} = \frac{1}{N}\sum_{i=1}^{N}(\phi^d w_t + \eta_t + u_{it})$$

$$L_{Et} = \frac{1}{N}\sum_{i=1}^{N}\phi^d w_t + \frac{1}{N}\sum_{i=1}^{N}\eta_t + \frac{1}{N}\sum_{i=1}^{N}u_{it}$$

Again, pulling constants out of the summations:

$$L_{Et} = \phi^d w_t\left(\frac{1}{N}\sum_{i=1}^{N}1\right) + \eta_t\left(\frac{1}{N}\sum_{i=1}^{N}1\right) + \frac{1}{N}\sum_{i=1}^{N}u_{it}$$

Since $\frac{1}{N}\sum_{i=1}^{N}1 = \frac{1}{N}\cdot N = 1$, and defining the equal-weighted idiosyncratic shock as $u_{Et} = \frac{1}{N}\sum u_{it}$, we get:

$$L_{Et} = \phi^d w_t + \eta_t + u_{Et}$$

Back

# GIV identifies supply elasticity

To find $\psi$, we use $z_t$ as an instrument for $L_{St}$ in the supply equation. We take the expectation of the product of the supply equation and the instrument:

$$\mathbb{E}[w_t z_t] = \mathbb{E}[(\psi L_{St} + \epsilon_t) z_t]$$
$$\mathbb{E}[w_t z_t] = \psi \mathbb{E}[L_{St} z_t] + \mathbb{E}[\epsilon_t z_t]$$

By the exogeneity condition, $\mathbb{E}[\epsilon_t z_t] = 0$. Thus:

$$\psi = \frac{\mathbb{E}[w_t z_t]}{\mathbb{E}[L_{St} z_t]} \tag{2}$$

This is the standard IV formula where $z_t$ instruments for $L_{St}$.

Back

# GIV identifies demand elasticity

To find $\phi^d$, we use the equal-weighted aggregate demand equation derived on slide 8:

$$L_{Et} = \phi^d w_t + \eta_t + u_{Et} \tag{41}$$

We aim to show that $\mathbb{E}[(L_{Et} - \phi^d w_t) z_t] = 0$. Substituting the demand equation:

$$\mathbb{E}[(L_{Et} - \phi^d w_t) z_t] = \mathbb{E}[(\eta_t + u_{Et}) z_t] = \mathbb{E}[\eta_t z_t] + \mathbb{E}[u_{Et} z_t] \tag{3}$$

We already know $\mathbb{E}[\eta_t z_t] = 0$. Now we examine $\mathbb{E}[u_{Et} z_t]$. Under the assumption that $u_{it}$ are i.i.d. with variance $\sigma_u^2$:

$$\begin{aligned} \mathbb{E}[u_{Et} z_t] &= \mathbb{E}[u_{Et}(u_{St} - u_{Et})] \\ &= \mathbb{E}[u_{Et} u_{St}] - \mathbb{E}[u_{Et}^2] \end{aligned}$$

Back

# GIV identifies demand elasticity

$$\mathbb{E}[u_{Et}u_{St}] - \mathbb{E}[u_{Et}^2] =$$

$$= \mathbb{E}\left[\left(\frac{1}{N}\sum_i u_{it}\right)\left(\sum_j S_j u_{jt}\right)\right] - \mathbb{E}\left[\left(\frac{1}{N}\sum_i u_{it}\right)^2\right]$$

$$= \frac{1}{N}\sum_i S_i \sigma_u^2 - \frac{1}{N^2}\sum_i \sigma_u^2$$

$$= \frac{1}{N}\sigma_u^2(1) - \frac{1}{N}\sigma_u^2 = 0$$

Since $\mathbb{E}[u_{Et}z_t] = 0$, it follows that $\mathbb{E}[(L_{Et} - \phi^d w_t)z_t] = 0$. Rearranging:

$$\mathbb{E}[L_{Et}z_t] = \phi^d \mathbb{E}[w_t z_t]$$

$$\phi^d = \frac{\mathbb{E}[L_{Et}z_t]}{\mathbb{E}[w_t z_t]}$$

# Simple GIV leaks macro shocks

The simple GIV is defined as $z_t^{simple} = L_{St} - L_{Et}$. Let's derive its components:

1. **Size-weighted average ($L_{St}$):**

$$\begin{aligned} L_{St} &= \sum S_i L_{it} = \sum S_i(\phi w_t + \lambda_i \eta_t + u_{it}) \\ &= \phi w_t \underbrace{\sum S_i}_{1} + \eta_t \underbrace{\sum S_i \lambda_i}_{\lambda_S} + \underbrace{\sum S_i u_{it}}_{u_{St}} \\ &= \phi w_t + \lambda_S \eta_t + u_{St} \end{aligned}$$

2. **Equal-weighted average ($L_{Et}$):**

$$\begin{aligned} L_{Et} &= \frac{1}{N} \sum L_{it} = \phi w_t + \eta_t \underbrace{\frac{1}{N} \sum \lambda_i}_{\lambda_E} + u_{Et} \\ &= \phi w_t + \lambda_E \eta_t + u_{Et} \end{aligned}$$

3. **The Resulting GIV:** Back

$$z_t^{simple} = L_{St} - L_{Et} = (\boldsymbol{\lambda_S - \lambda_E})\boldsymbol{\eta_t} + (u_{St} - u_{Et})$$

# Econometrics Refresher

## OLS and the Annihilator Matrix

Standard OLS projections decompose a vector $y$ into explained and unexplained parts:

- **Projection Matrix ($P$):** $P = X(X'X)^{-1}X'$. Projects $y$ onto the column space of $X$. $\hat{y} = Py$.
- **Annihilator Matrix ($M$):** $M = I - P$. Projects $y$ onto the space orthogonal to $X$.
- **Orthogonality:** $MX = 0$. This ensures the residuals $e = My$ are uncorrelated with the regressors $X$.
- **Residuals:** $e = y - \hat{y} = (I - P)y = My$.

*Notation Map:*
Slide $\lambda \equiv$ Standard $X$ (Regressors/Factors)
Slide $Q \equiv$ Standard $M$ (Residual Maker Matrix)
Slide $S \equiv$ Standard $y$ (Data Vector)
Slide $\Gamma \equiv$ Standard $e$ (Residuals/Granular Weights) Back

# Appendix B

Propositions

# Orthogonal weights isolate idiosyncratic shocks

The GIV $z_t$ is constructed from observables, $L_{it}$, using weights $\Gamma \in \mathbb{R}^N$ orthogonal to factor loadings:

$$z_t := \Gamma' L_t = \sum_{i=1}^{N} \Gamma_i L_{it}$$

By construction, this isolates a linear combination of idiosyncratic shocks $u_t$:

$$z_t = \Gamma' u_t \quad \Rightarrow \quad \text{exogenous to aggregate shocks } \eta_t, \varepsilon_t$$

This allows us to construct the sample estimators:

$$\hat{\psi}_T = \frac{\sum_{t=1}^{T} w_t z_t}{\sum_{t=1}^{T} L_{St} z_t} \qquad \hat{\phi}_T^d = \frac{\sum_{t=1}^{T} L_{\tilde{E}t} z_t}{\sum_{t=1}^{T} w_t z_t}$$

Back

# Moments identify elasticities

The estimators follow directly from the exogeneity of $z_t$ to structural shocks:

**Inverse Supply Elasticity**

From $w_t = \psi L_{St} + \cdots + \varepsilon_t$:

$$\mathbb{E}[(w_t - \psi L_{St}) z_t] = 0$$

$$\Rightarrow \quad \psi = \frac{\mathbb{E}[w_t z_t]}{\mathbb{E}[L_{St} z_t]}$$

**Demand Elasticity**

For weights $\tilde{E}$ where $\mathbb{E}[u_{\tilde{E}t} u_{\Gamma t}] = 0$:

$$\mathbb{E}[(L_{\tilde{E}t} - \phi^d w_t) z_t] = 0$$

$$\Rightarrow \quad \phi^d = \frac{\mathbb{E}[L_{\tilde{E}t} z_t]}{\mathbb{E}[w_t z_t]}$$

Back

# Proposition 1

Under standard regularity conditions, as the number of time periods $T \to \infty$:

**Inverse Supply Elasticity**

$$\hat{\psi}_T \xrightarrow{\text{a.s.}} \psi$$

**Demand Elasticity**

$$\hat{\phi}^d_T \xrightarrow{\text{a.s.}} \phi^d \quad (\text{if } \psi \neq 0)$$

**Intuition:** The orthogonal weighting $\Gamma$ ensures the instrument is uncorrelated with the structural error term, resolving simultaneity bias.

Back

# Multipliers capture amplification

To understand the variance of these estimators, we define two key multipliers:

**Quantity Pass-through**

$$M := \frac{1}{1 - \psi\phi^d}$$

**Price Pass-through**

$$\mu := \psi M$$

These represent the pass-through from idiosyncratic and aggregate shocks to the aggregate quantity and the price, respectively.

Back

# Proposition 2

The estimators converge in distribution to a normal distribution:

$$\sqrt{T}(\hat{\psi}_T - \psi) \xrightarrow{d} \mathcal{N}(0, \sigma_\psi^2), \quad \sigma_\psi = \frac{\mathbb{E}[u_{\Gamma t}^2]^{1/2}}{|\mathbb{E}[u_{St} u_{\Gamma t}]|} \frac{\sigma_{\varepsilon\psi}}{|M|}$$

$$\sqrt{T}(\hat{\phi}_T^d - \phi^d) \xrightarrow{d} \mathcal{N}(0, \sigma_{\phi^d}^2), \quad \sigma_{\phi^d} = \frac{\mathbb{E}[u_{\Gamma t}^2]^{1/2}}{|\mathbb{E}[u_{St} u_{\Gamma t}]|} \frac{\sigma_{\varepsilon\phi}}{|\mu|}$$

Back

# Volatility improves precision

The variance equations reveal the signal-to-noise ratio inherent to GIV:

**Signal Strength**

$|\mathbb{E}[u_{St}u_{\Gamma t}]|$

Larger granular shocks increase relevance.

**Instrument Noise**

$\mathbb{E}[u_{\Gamma t}^2]^{1/2}$

Variance of the constructed GIV itself.

**Structural Noise**

$\sigma_\varepsilon/|M|$

Unobserved shocks dampening precision.

Back

# Proposition 3

To minimize the asymptotic variances $\sigma^2_\psi$ and $\sigma^2_{\phi^d}$, we use the $N \times N$ projection matrix $Q$ that is orthogonal to the factor loadings $\lambda$:

$$Q := I - \lambda(\lambda'\lambda)^{-1}\lambda'$$

The optimal weights vector $\Gamma^*$ is the projection of the size vector $S$:

$$\Gamma^* = S'Q$$

**Result:** $\Gamma^*$ is the vector closest to $S$, while remaining orthogonal to the unobserved aggregate factor loadings.